

Industrial operations intelligence for a five-plant Indian manufacturing company.

YantraMitra unifies plants, machines, sensors, work orders, digital twins, and AI agents into a single operating layer so teams can understand issues and act quickly.

5

Facilities

29

Machines

174

Sensors monitored

Scenario

The seeded company operates Pune automotive components, Ahmedabad textile and chemical process lines, Chennai electronics assembly, Bengaluru precision engineering, and a Nagpur logistics hub. Each plant has plant-specific machines, risks, and operational workflows.

Pune Automotive

Robotic welding, CNC cells, AGV movement, and drivetrain component production.

Ahmedabad Process

Textile finishing, dye tanks, process parameters, and chemical quality context.

Chennai Electronics

SMT machines, AOI inspection, reflow ovens, and electronics test benches.

Bengaluru Precision

Micro-machining, metrology, CMM inspection, calibration, and additive manufacturing.

Nagpur Logistics

ASRS aisles, sortation, dock operations,
AGVs, and warehouse flow monitoring.

What the app contains

Command Center

Live KPIs, open incidents, critical alerts, facility health cards, and agent activity.

Global Plant Map

All five plants pinned using their real Indian cities with cards and drilldowns.

Plant Overview

A dynamic plant route renders the requested facility instead of hardcoded city data.

Digital Twin

Machine state, sensor telemetry, and fault context are visualized together.

Asset Fleet

Machine inventory, health, alert state, and detail routes for asset-level inspection.

YantraNkian AI Console

A plant-aware assistant for asking operational questions and drafting next steps.

Agents

Mission control for diagnostic, planner, sentinel, and operations agents.

Work Orders

Maintenance execution with assignment, status, approval, and closure tracking.

Signal-to-action loop

01 Sense

02 Understand

03 Recommend

04 Approve

05 Execute

The core design goal is continuity. Operators should not lose context when moving from an alert to a machine, from a machine to an agent explanation, or from an explanation to a work order.

Deployment and operating model

Authentication and roles

Public pages explain the platform. Application pages require authentication. Signup supports role selection, and the app separates manager, maintenance, operator, and executive workflows.

Data model

The seeded database contains plants, machines, sensors, alerts, work orders, agent missions, and profile data. The public home page uses the same counts so the landing experience feels connected to the actual product.

AI assistant

YantraNkian can be connected to OpenAI with a company-owned API key. For production, configure secrets in environment variables, restrict access by role, and keep human approval for physical maintenance actions.

Release checklist

- Use a production JWT secret and environment-managed API keys.
- Connect the database to company-approved infrastructure.
- Validate agent recommendations through plant operating procedures.
- Confirm sensor integrations and work-order ownership rules before live use.
- Keep public documentation accurate as plant data or workflows change.